CS 532, Spring 2020: Quiz 1

NAME:

Problem 1. Let $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ denote three points on the projective plane. Using ONLY dot and cross products, write an expression that test/evaluates whether or not 3 points are collinear (i.e. they are all on the same line). Under what scenario (if any) would your test fail?

Problem 2. Let $\mathbf{X}$ and $\mathbf{x}$ denote, respectively, the homogeneous coordinates of a 3D point and the homogeneous 2D coordinates of a point in an image plane. Answer the following:

1. What dimensions (i.e. rows and columns) is the projection matrix $\mathbf{M}$ that maps $\mathbf{X}$ into $\mathbf{x}$.

2. Write the explicit inhomogenous 2D coordinates of the point $\mathbf{x}$ in terms of the entries m_{ij} of the matrix $\mathbf{M}$ determined above and the coordinates of $\mathbf{X}$

Problem 3. Let ℓ_1 and ℓ_2 denote a pair of distinct vertical 2D lines, each one separated d units from the 2D origin, while ℓ_3 is a horizontal line passing through the origin.

1. Write the line coordinates for all three lines

2. How many intersection points can be determined among them?